2024 VCE Specialist Mathematics 2 (NHT) external assessment report

Specific information

This report provides sample answers or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

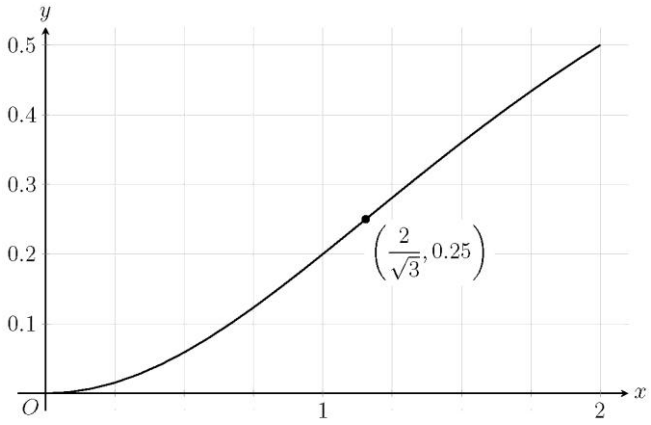
Section A – Multiple-choice questions

Question	Correct answer
1	C
2	E
3	E
4	D
5	A
6	D
7	E
8	B
9	A
10	C
11	D
12	E
13	B
14	B
15	D
16	D
17	A
18	B
19	C
20	A

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Section B – Extended response questions

Question 1a.



Question 1bi.

$$\pi\int_0^{0.5}\left(\frac{4}{1-y}-4\right)dy, \text{ or } \pi\int_0^{0.5}\frac{4y}{1-y}dy$$

Question 1bii.

$$2\pi\left(2\log_e(2)-1\right)$$

Question 1c.

$$2\pi\int_0^{0.5}\sqrt{\frac{4y}{1-y}}\sqrt{1+\left(\frac{d}{dy}\sqrt{\frac{4y}{1-y}}\right)^2}\,dy, \text{ or } 2\pi\int_0^{0.5}\sqrt{\frac{4y}{1-y}}\sqrt{1+\frac{1}{y(1-y)^3}}\,dy, \text{ or } 2\pi\int_0^{2}x\sqrt{1+\left(\frac{8x}{\left(x^2+4\right)^2}\right)^2}\,dx$$

$$=13.08$$

Question 1d.

$$\frac{dy}{dt}=\frac{dV}{dt}\times\frac{dy}{dV}$$

$$\frac{dV}{dy}=\frac{4\pi y}{1-y}$$

$$\frac{dy}{dt}=0.1\times\left(\frac{1-y}{4\pi y}\right)$$

$$\frac{dy}{dt}=0.1\times\left(\frac{1-0.25}{4\times\pi\times0.25}\right)=0.024$$

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Question 2a.

$$\begin{aligned}\text{LHS} &= \left(\cos\left(n\theta\right)+i\sin\left(n\theta\right)\right)\times\left(\cos\theta+i\sin\theta\right)\\ &= \cos\left(n\theta\right)\cos\left(\theta\right)+i\sin\left(n\theta\right)\cos\left(\theta\right)+\cos\left(n\theta\right)i\sin\left(\theta\right)-\sin\left(n\theta\right)\sin\left(\theta\right)\\ &= \cos\left(n\theta\right)\cos\left(\theta\right)-\sin\left(n\theta\right)\sin\left(\theta\right)+i\left(\sin\left(n\theta\right)\cos\left(\theta\right)+\cos\left(n\theta\right)\sin\left(\theta\right)\right)\\ &= \cos\left(n\theta+\theta\right)+i\sin\left(n\theta+\theta\right)\\ &= \cos\left(\left(n+1\right)\theta\right)+i\sin\left(\left(n+1\right)\theta\right)\\ &= \text{RHS}\end{aligned}$$

Question 2b.

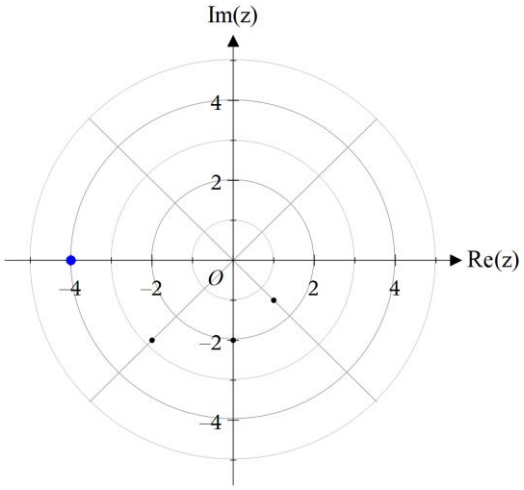
$$\left(\sqrt{2}\right)^{n}\text{cis}\left(-\frac{n\pi}{4}\right)$$

Question 2c.

$-2-2i$

Question 2d.

Point $\left(-4,0\right)$ plotted.



Question 2e.

$$-\frac{n\pi}{4}=-2\pi k \qquad \text{or,} \qquad -\frac{n\pi}{4}=2\pi k$$

$$n=8k \text{ where } k\in Z^{+} \qquad n=-8k \text{ where } k\in Z^{-}$$

or $8, 16, 24, \ldots$

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Question 2f.

$a = 2, \quad b = -2$

Question 2g.

$$\text{Angle} = 2 \times \cos^{-1}\left(\frac{\sqrt{2}}{\sqrt{10}}\right) \approx 2.2143$$

$$\text{Area} = \frac{1}{2} \times 10 \times \left(2.2143 - \sin\left(2.2143\right)\right) = 7.07$$

Question 3a.

$V = v(8) = 25$

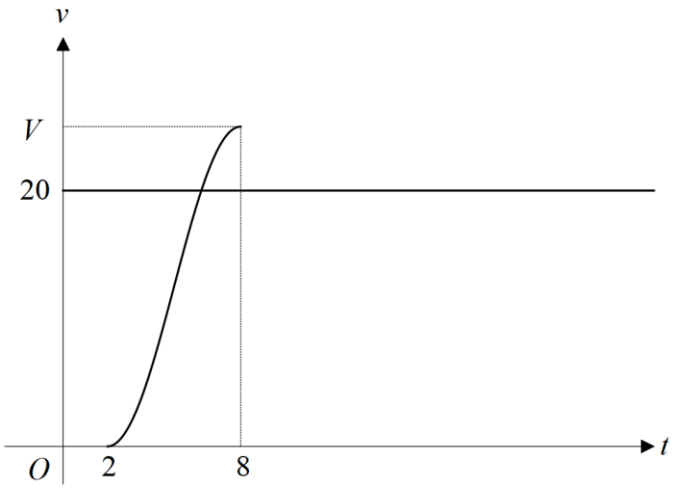
Question 3bi.

$$a(t) = \frac{25\pi}{12} \sin\left(\frac{\pi}{6}(t-2)\right), \text{ or } a(t) = -\frac{25\pi}{12} \cos\left(\frac{\pi}{6}(t+1)\right)$$

Question 3bii.

6.5

Question 3c.



Question 3d.

$20t$

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Question 3e.

$$\int_{2}^{8} 12.5 - 12.5\cos\left(\frac{\pi}{6}(t-2)\right) dt = 75$$

Question 3f.

$$75 + (t-8) \times 25 = 20t$$

$$t = 25$$

Question 3g.

$$20 \times 25 = 500$$

Question 4ai.

$$\frac{3t}{5} = 30$$

$$\left(t = \frac{30 \times 5}{3}\right)$$

$$t = 50$$

Question 4aii.

$$\tan(\theta°) = \frac{30}{\sqrt{\left(50 + 25\sin\left(\frac{50\pi}{30}\right)\right)^2 + \left(50 + 25\cos\left(\frac{50\pi}{30}\right)\right)^2}}$$

$$\theta = 24$$

Question 4b.

$$\underset{\sim}{r}(0) = 50\underset{\sim}{i} + 75\underset{\sim}{j}$$

$$50 + 25\sin\left(\frac{\pi t}{30}\right) = 50 \qquad 50 + 25\cos\left(\frac{\pi t}{30}\right) = 75$$

$$t = 30, 60, 90, \ldots \qquad t = 60, 120 \ldots$$

$$\Rightarrow t = 60$$

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Question 4c.

$$\dot{\underset{\sim}{r}}(t) = \frac{25\pi}{30}\cos\left(\frac{\pi t}{30}\right)\underset{\sim}{i} - \frac{25\pi}{30}\sin\left(\frac{\pi t}{30}\right)\underset{\sim}{j} + \frac{3}{5}\underset{\sim}{k}$$

$$\ddot{\underset{\sim}{r}}(t) = -\frac{25\pi^2}{30^2}\sin\left(\frac{\pi t}{30}\right)\underset{\sim}{i} - \frac{25\pi^2}{30^2}\cos\left(\frac{\pi t}{30}\right)\underset{\sim}{j}$$

$$\dot{\underset{\sim}{r}}(t).\ddot{\underset{\sim}{r}}(t) = -\frac{25^2\pi^3}{30^3}\cos\left(\frac{\pi t}{30}\right)\sin\left(\frac{\pi t}{30}\right) + \frac{25^2\pi^3}{30^3}\sin\left(\frac{\pi t}{30}\right)\cos\left(\frac{\pi t}{30}\right)$$

$$= 0$$

$\therefore \dot{\underset{\sim}{r}}(t)$ is perpendicular to $\ddot{\underset{\sim}{r}}(t)$ for all t.

Question 4d.

$$\left|\dot{\underset{\sim}{r}}(t)\right| = \sqrt{\left(\frac{25\pi}{30}\right)^2 + \left(\frac{3}{5}\right)^2} = 2.69$$

Question 4e.

$$\underset{\sim}{r}(t) - \underset{\sim}{r}_{\text{tree}} = \left(50 + 25\sin\left(\frac{\pi t}{30}\right) - 60\right)\underset{\sim}{i} + \left(50 + 25\cos\left(\frac{\pi t}{30}\right) - 25\right)\underset{\sim}{j} + \left(\frac{3t}{5} - 8\right)\underset{\sim}{k}$$

$$\left|\underset{\sim}{r}(t) - \underset{\sim}{r}_{\text{tree}}\right| = \sqrt{\left(25\sin\left(\frac{\pi t}{30}\right) - 10\right)^2 + \left(25 + 25\cos\left(\frac{\pi t}{30}\right)\right)^2 + \left(\frac{3t}{5} - 8\right)^2}$$

Minimum value of $\left|\underset{\sim}{r}(t) - \underset{\sim}{r}_{\text{tree}}\right| = 7.9$

Question 5a.

$\overrightarrow{AB} = \underset{\sim}{i} + 2\underset{\sim}{j} + \underset{\sim}{k}, \quad \overrightarrow{BC} = 3\underset{\sim}{i} - \underset{\sim}{j} - 4\underset{\sim}{k}$

$\overrightarrow{AB}$ is not parallel to $\overrightarrow{BC}$ $\left(\overrightarrow{AB} \neq m\overrightarrow{BC}\text{, where } m \text{ is a non-zero real number}\right)$

A, B, C are not collinear.

Other combinations of vectors are suitable. Other correct approaches could include determining magnitudes of vectors, cross product or scalar product.

Question 5b.

A normal vector, $\underset{\sim}{n} = -7\underset{\sim}{i} + 7\underset{\sim}{j} - 7\underset{\sim}{k}$

$-7x + 7y - 7z = d$

$A(-1, 1, 2)$

$-7(-1) + 7(1) - 7(2) = d$

$d = 0$

$-7x + 7y - 7z = 0$ (or any scalar multiple)

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Question 5c.

$x = 1 + 2t, \quad y = -2 + t, \quad z = 3 - t$

$2(1 + 2t) - (-2 + t) + 3(3 - t) = 13 \neq 1$

$\therefore$ no point of intersection

Question 5d.

A normal vector to the plane, $\underset{\sim}{n} = 2\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k}$

$(0, -1, 0)$ lies on the plane, $(1, -2, 3)$ lies on the line.

A vector from line to plane is $\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k}$

$$\text{Distance} = \left| \left(\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k} \right) . \frac{2\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k}}{\sqrt{14}} \right| = \frac{12}{\sqrt{14}}.$$

Question 6a.

$$\text{E}(\bar{X}) = 0.37, \quad \text{sd}(\bar{X}) = \frac{0.12}{\sqrt{9}} = 0.04$$

Question 6b.

$$1.645 \times \frac{0.12}{\sqrt{n}} \leq 0.05$$

$$n > 15.58 \Rightarrow n = 16$$

Question 6c.

$H_0 : \mu = 0.37, \quad H_1 : \mu > 0.37$

Question 6di.

$$p = \Pr(\bar{X} > 0.41 \mid \mu = 0.37) = 0.006$$

Question 6dii.

$p < 0.01 \Rightarrow$ reject H_0

(mean is greater than 0.37)

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Question 6e.

$\mathrm{E}\left(\bar{X}\right) = 0.37, \quad \mathrm{sd}\left(\bar{X}\right) = \dfrac{0.12}{\sqrt{56}}$

$\Pr\left(\bar{X} > \bar{x}\right) = 0.01$

$\bar{x} = 0.407$

Question 6f.

$\mathrm{sd}\left(\bar{X}\right) = \dfrac{0.12}{\sqrt{56}}$

$\Pr\left(\bar{X} > \bar{x} \mid \mu = 0.37\right) = 0.05$

$\bar{x} = 0.396\ldots$

$\Pr\left(\bar{X} < 0.396\ldots \mid \mu = 0.40\right) = 0.4$

So Pr(Type II error) is 0.4

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